Eduardo Mendes

PhD Electrical & Computer's Engineering

EMBEDDED SYSTEMS / HAPTICS / ACOUSTICS

Eduardogilmendes@gmail.com

1992, Braga (Portugal)

PhD in Electrical and Computers Engineering from University of Minho. **Embedded Systems** and **Haptics** Specialities.

Collaboration in innovation projects in industrial setting (as embedded developer) such as **Active Noise Cancelling** and **Ultrasonic-Haptics**.

Interest for Mathematics, Acoustics and Electronics. Experience in embedded development, FPGA development, Matlab simulation and Acoustic Measurements.

Programming Languages:
C/C++; Python; Matlab; Verilog;

Programming Tools:
Matlab; Spyder; Vivado; VS Code;

Technologies and Architectures:
Xilinx Zynq-7000; ARM Cortex-A9; ARM TrustZone.

Other

Erasmus in Thainland
Languages: Portuguese, English, Spanish

Experience

PhD Electrical & Computer's Eng

2020-2026 University of Minho

Theme: "Optimizing Spatio-Temporal Feedback for Automotive **Ultrasonic Mid-Air Haptics**"

The goal of the PhD was to **optimized haptic sensations** produced by UMAH systems.
To that end, user experiments for haptic sensation parameterization were conducted. Then, a haptic sensation generation tool was developed. It abstracts the UX designer from the parametric details and automatic creates the sensations. Novel haptic sensations were created, that can be used in UXes. Acoustic simulation in Matlab (as well as in Python) were performed for stimuli validation. Moreover, extensive acoustic testing was performed.

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Researcher Engineer

2018-2020 Guimarães

Researcher on the INOVATIVE Car HMI project. I've been involved in the development of prototypes and demonstrators for **Active Noise Cancelling** and **Haptics**.
Work consisted in improving and porting the system to new platform boards that were tested in vehicles.
Xilinx Zynq boards, C/C++ , FPGA and Matlab

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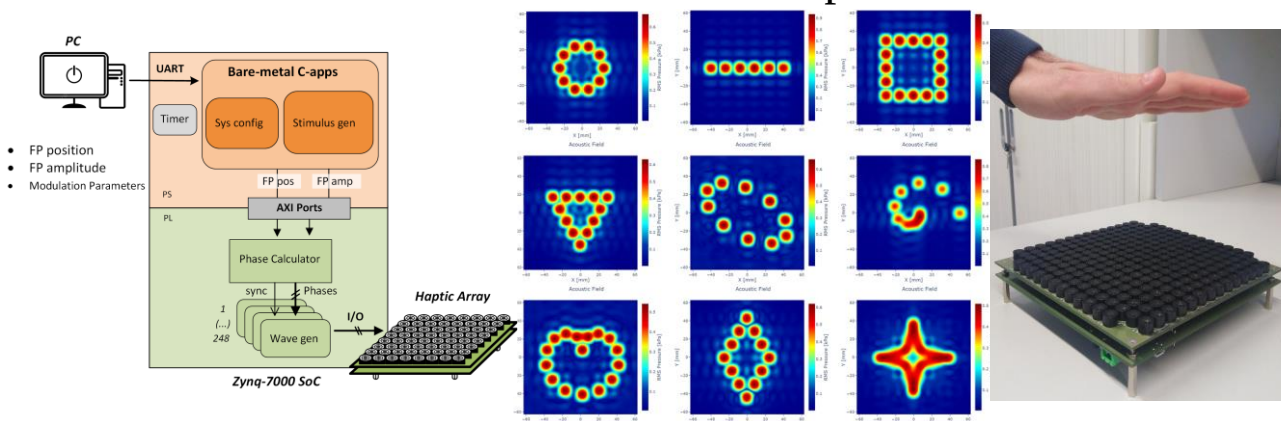
MSc Electrical & Computer's Eng

2011-2017 University of Minho

Integrated master in electrical and computers engineering. Courses on **Mathematics**, **Physics** and **Electronics**. Master specialization in **Embedded Systems** and **Power Systems**.
Dissertation: "Extending security spectrum of ARM TrustZone" 18/20

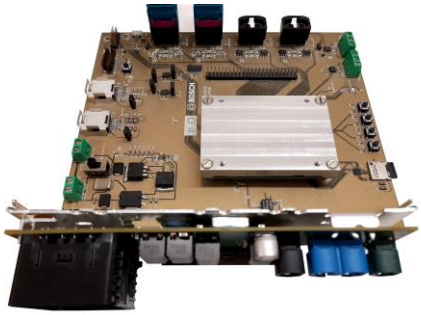
Projects

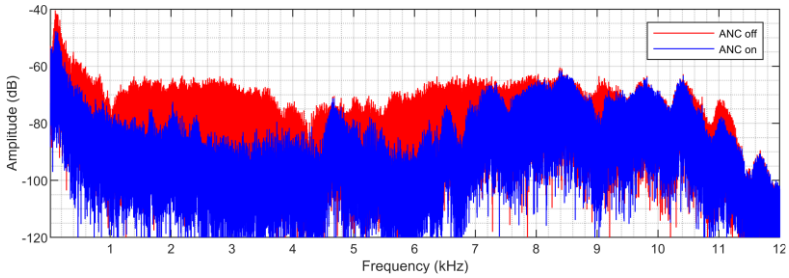
Ultrasonic Mid-Air Haptics

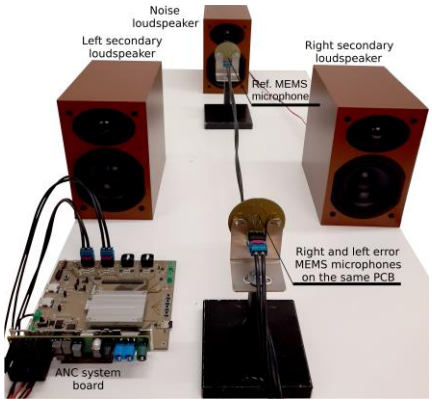


The Ultrasonic Mid-Air Haptic developed during PhD. The left image depicts an overview of the system's architecture (coded in C and Verilog). The figure in the middle shows haptic stimuli created by the Haptic Sensation Tool (implemented in Matlab). And on the right image, one can observe the UMAH prototype.

Active Noise Cancellation







Active Noise Cancellation was one of the projects I was involved.
The pair of images of the left shows the ANC prototype (alone and in a demonstration setup). The top image of the right show the performance of the ANC system. In other words, the graphic shows the attenuation of the system of broad band pink noise.

Trossingen Lyre

The Trossingen Lyre is personal project. It reflects my interest in music, acoustic and handcrafting. I was able to build a sounding instrument. Together with other musicians we did public free public shows showcasing the instrument and playing ancient music.